

Colombia

Superintendence of Industry and Commerce

I. Powers of the Superintendence of Industry and Commerce in Matters of Free Economic Competition

The Superintendency of Industry and Commerce (hereinafter SIC), through the Deputy Superintendency for the Protection of Competition, has as its main role the promotion and protection of free economic competition through the exercise of both *ex ante* and *ex post* functions.

On the one hand, the Deputy Superintendency carries out *ex ante* or preventive control through the Competition Advocacy, Business Concentrations, and Compliance Directorate Work Groups. The first aims to prevent restrictions on competition in the market through the review of draft administrative acts that regulatory authorities intend to issue. The second is responsible for analyzing business integration operations that may have an impact on free competition. Finally, the Compliance Directorate undertakes, among other actions, dissemination, promotion and training activities related to fostering and building a culture of compliance in terms of free economic competition. Additionally, this Directorate also supports companies in the effective adoption of compliance programmes.

Overall, *ex ante* control is centered on the implementation of preventive measures designed to promote a culture of compliance with free competition regulations in Colombia. Through these measures, the SIC seeks to foster greater awareness among businesses and consumers regarding the significance and advantages of adhering to free competition policies, while also safeguarding open market participation, consumer welfare, and economic efficiency.

On the other hand, there is *ex post* or corrective control, which is carried out through the Restrictive Practices, Competition Promotion and Protection, and Elite Anti-Collusion Work Groups. This corrective control is conducted when the Authority has identified the possible occurrence of behaviors that restrict free economic competition in the market. In this regard, these groups are responsible for carrying out administrative investigations to determine whether or not actions contrary to the free competition protection regime have been committed, including, for example, anti-competitive agreements or abuse of dominant market position.

II. Computational Activities Related to the Collection, Preservation, and Analysis of Digital Evidence

SIC considers it important to highlight the computational activities aimed at the collection, preservation and analysis of digital evidence related to potential anti-competitive practices. Specialized tools are used to extract information from devices (computers, emails, mobile phones and/or tablets) identified during unannounced inspections (dawn raids) carried out by SIC. Once the information has been collected, SIC uploads the data into specialized licensed software used to access the content and search for keywords or terms that can link evidence used to verify potentially anti-competitive conduct. It is important to note that this software facilitates the visualization of chats, emails, images, and other types of files.

To illustrate the computational process related to the collection, processing, and analysis of various digital evidence, the following elements are presented.

A. Stage 1—Acquisition of Digital Evidence

During unannounced inspections (dawn raids), IT specialists undertake the acquisition of digital evidence through data copying or forensic imaging techniques. These procedures are conducted in accordance with established protocols to safeguard the integrity of the information. To ensure evidentiary reliability, a documented chain of custody is maintained, and cryptographic hash values are generated to validate the authenticity and immutability of the data vis-à-vis its original source.

B. Stage 2—Processing and Secure Preservation

Upon acquisition, the digital evidence is subjected to a structured processing phase. The information is securely stored and access is strictly limited to authorized personnel directly assigned to the case. Any request for access must be formally justified and logged. This stage ensures the preservation of evidential integrity and prepares the data for subsequent forensic analysis.

C. Stage 3—Forensic Analysis

Guided by the lead investigator, forensic analysts apply digital filtering and analytical tools to identify data relevant to the working hypothesis of the case. The evidence is categorized using metadata tags, enabling targeted review and thematic

classification. The preliminary findings undergo peer validation within the investigative team to ensure coherence, legal robustness, and procedural fairness.

III. Projects Related to Technological Tools that Support Data Processing and Analysis Tasks

Additionally, SIC has implemented projects related to technological tools aimed at: (i) improving the capacity to detect potentially anti-competitive behaviors; (ii) increasing efficiency in examining large volumes of structured and unstructured data; and (iii) better allocation of human resources. Currently, these tools include: *Sabueso*, *Inspector*, *Sherlock*, and *Búho*. As SIC expands its monitoring capabilities and refines its methods for detecting structural and behavioral anomalies, the likelihood of timely actions to protect and promote free economic competition increases.

A. SABUESO

This tool was developed to monitor the pricing behavior of products offered by major supermarket chains, including Éxito, Carulla, Metro, Olímpica, and Jumbo. It operates by scraping price data from the supermarkets' websites, which are then normalized and stored with information on product presentation, quantity, category, subcategory, and price. The architecture relies on microservices built in Python, a PostgreSQL database, and cloud services hosted on Microsoft Azure. Although the tool lacks a graphical user interface, the data are analyzed either manually or via Power BI dashboards.

B. BÚHO

Búho is designed to support and facilitate the identification of news articles that could indicate potential breaches of the competition regime. The tool scrapes headlines from digital media outlets such as Las2orillas, La W, and RCN News using pre-established keywords. These headlines are then cleaned, filtered, and compared with a database to avoid duplication. The final selection is compiled and sent in weekly notifications via email to designated officials within the Deputy Superintendency for the Protection of Competition. Búho makes use of Selenium, PostgreSQL, and natural language processing (NLP) techniques.

C. INSPECTOR

Inspector aims to monitor regulatory projects initiated by 64 public entities that may have an impact on free economic competition. It extracts documents from government websites using ChromeDriver and XPath, filtering them through

keywords and summarizing content to generate metadata. The tool sends daily email notifications to the Competition Advocacy Work Group, including improved titles, deadlines for comments, and direct access to attached regulatory projects. The technological backbone comprises Python, PostgreSQL, NLP, Microsoft Azure, and Kubernetes.

D. SABUESO—Flights

A complementary version of Sabueso was designed to monitor airline ticket prices offered by carriers and metasearch engines such as Despegar, Viajala, and eDestinos. The tool captures prices for direct domestic flights with hand luggage included. It filters out non-direct flights and normalizes the data (airline, departure time, price). Outliers are flagged, and alerts are issued monthly. This tool also employs Selenium, ChromeDriver, PostgreSQL, and Azure. Nevertheless, it has faced operational challenges such as CAPTCHA barriers, IP restrictions, and currency changes by the metasearch providers.

E. SHERLOCK

Sherlock was developed to reduce the time required for mechanical reviews carried out by the Elite Anti Collusion Work Group, using data from the SECOP I, SECOP II, and Open Data in Colombia public procurement platforms via the Socrata API. The tool applies filters to identify tenders with single bidders, low competition, or minimal discounts—indicators of potential collusion. Alerts are generated biweekly and shared in Excel reports. It utilises Python, PostgreSQL, and Azure, incorporating automatic data validations. However, Sherlock does not detect deserted processes, non-awarded contracts, or complex plural structures, and occasionally issues alerts for legally expired tenders.

These tools exemplify how the Deputy Superintendency for the Protection of Competition has integrated computational capabilities into both its daily workflows and specialized missions. All tools have been developed with specific objectives in mind, using advanced programming techniques and, in some cases, artificial intelligence.

Institutional challenges persist. Limitations in budget and human talent remain, particularly due to the continuing need for manual data interpretation, as full automation has yet to be achieved. Furthermore, coordination across institutions poses an obstacle, especially for tools such as Inspector and Búho, which rely on consistent keyword updates and maintenance of diverse data sources.

From a technical perspective, while the use of modern technologies such as microservices, containers, and cloud computing has enabled significant progress, the absence of user-friendly graphical interfaces continues to hinder accessibility for non-technical users. Sabueso-Flights in particular has been affected by CAPTCHA restrictions and currency inconsistencies, and Sherlock is limited by its reliance on the timely updating of public datasets.

Despite these difficulties, the tools have contributed to the internal transformation of the institution. Manual data-gathering processes have been significantly automated, leading to more efficient analysis. Historical data repositories have been established through tools such as Inspector and Sherlock, facilitating longitudinal studies on regulatory and procurement trends. The adoption of cloud-based and modular architectures reflects a broader commitment to technological modernization.

The institution has set ambitious goals, including the consolidation of early warning systems to detect competition risks, the integration of data visualization dashboards to support decision-making, and the expansion of data coverage to include more entities, media outlets, and product categories. It also seeks to enhance the detection of plural structures in public procurement tenders, thereby improving the effectiveness of Sherlock.

From these experiences, several lessons have been drawn. Firstly, the automation of data collection is insufficient on its own; effective analysis and visualization tools are also required. Secondly, the collaboration between technical and legal teams is crucial for the success of these initiatives. Data normalization is a vital step in ensuring analytical quality, and human validation remains essential to avoid false positives. Moreover, the interoperability and standardization of systems and datasets have proven fundamental to maximizing analytical effectiveness and traceability.

Nonetheless, obstacles persist. The lack of graphical interfaces limits the reach of these tools. In the case of Inspector, the high volume of notifications without adequate filtering has sometimes generated noise. Meanwhile, despite automated data collection, analysis remains labor-intensive, particularly in tools such as Sherlock and Sabueso-Flights, which are still vulnerable to technical blocks and limited coverage due to third-party policy changes.

In conclusion, the development and deployment of the SABUESO, BÚHO, INSPECTOR, and SHERLOCK tools constitute significant advances in the use of data

analytics by the Superintendence of Industry and Commerce. Their potential impact could be further enhanced by improving user interfaces, integrating artificial intelligence for automated analysis, and promoting greater interoperability among systems. These experiences offer valuable insights for other jurisdictions seeking to strengthen their monitoring and competition advocacy capabilities.

Tool	Purpose	Data Source	Technology Stack
SABUESO (Retail)	Price monitoring in supermarkets	Éxito, Carulla, Metro, Olímpica, Jumbo	Python 3.11, PostgreSQL, Azure, Kubernetes
SABUESO (Flights)	Monitoring domestic airfare prices	Despegar, Viajala, eDestinos	Python 3.11, PostgreSQL, Azure, Selenium, ChromeDriver
SHERLOCK	Detects collusion in public procurement	SECOPI I & II, Open Data Colombia	Python 3.11, PostgreSQL, Azure, Socrata API
INSPECTOR	Alerts on anti-competitive regulatory proposals	69 public entities	ChromeDriver, XPath, PostgreSQL, Azure, Kubernetes
BÚHO	Captures relevant media content	La W, RCN Radio, Las2Orillas	Selenium, PostgreSQL, NLP, Azure